

				نموذج تقديم اعتماد مواد			
المشروع		توريد وتركيب وتشغيل أنظمة الإطفاء وإنذار الحريق لأحد المواقع التابعة لديوان رئاسة أمن الدولة بحافظة الطائف		قيمة العقد		1.375.641.5	
موقع المشروع		الطائف		المالك		رئاسة أمن الدولة	
مستشاري		مكتب نسج الابداع للاستشارات الهندسية		المقاول		باسكو سيفتي	
بيانات التقديم							
رقم التقديم		MAS-MECH-11-R00		تاريخ التقديم		29/04/2026	
التصنيف		معماري		مديني		ميكانيكا	
نوع التقديم		مستند		مواد		مخططات	
الكوود		الوصف		الكود		التسلسل	
1		Fire Pump Test Line (6")		10			
2		Supplier: Mechanical Future (مستقبل الميكانيكا)		11			
3		1: Gate Valve, 6", OS&Y, LEDE Brand (UL/FM) – Quantity: 2		12			
4		2: Flow Meter, 6", GVI Brand (UL/FM) – Quantity: 1					
5		3: Pressure Gauge, Reliable Brand – Quantity: 1					
6							
7							
8							
9							
الاستلام		المقاول		ممثل المقاول		التوقيع	
				م. احمد علي مصطفي		التاريخ	
						29/04/2026 م - 12/11/1447 هـ	
المراجعة الفنية		الملاحظات الفنية					
☐ المدني / الانشائي		لا مانع من استخدام المواد المرفقة أعلاه وحسب الكاتالوج المرفقة أدناه مع مراعاة تعليمات الشركة المصنعة. ركود البناء المودرن NFPA. تسليمات الجهة المالكة والاستشارية. يجب على المقاول قبل بدء التوريد أو التركيب التأكد من الاختبارات المذكورة بالكاتالوجات المرفقة وفي حال عدم صحتها يرجع للاستشارة والجهة المالكة.					
☐ المعماري							
☐ الكهربائي							
☒ الميكانيكي							
مدير المشروع							
التوقيع							
التاريخ		9/5/2026					
☐ مرفوض ويعاد التقديم: C		☒ مقبول حسب الملاحظات: B		☐ مقبول: A		الكود	
						العدد	
						15	
						المرفقات	
						التاريخ	
						ممثل المالك	

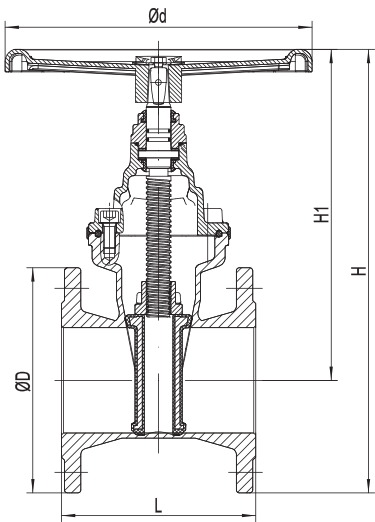
يتعهد المقاول المذكور أعلاه بأنه قد راجع كافة تفاصيل التقديم وطابقها مع شروط ومواصفات ومخططات المشروع المعتمدة وأن هذه الموافقة لا تخلي مسؤوليته حسب متطلبات العقد

DIN F4 Resilient seated NRS gate valve-flange end

Model No. Z45X-01



Name of parts	Material
Body	Ductile iron
Wedge	Ductile iron+EPDM
Stem nut	Bronze
Stem	SS431 or SS304, SS316, bronze
Bonnet	Ductile iron
Bonnet Gasket	EPDM
Sealing Ring	EPDM
Upper cover	Ductile iron
Sliding disc	Bronze
Screw	Carbon steel with hot galvanization
O-rings	EPDM
Handwheel	Ductile iron



Technical data >

- Sizes: DN50~DN300
- Working pressure: PN16/PN25
- Valve standard: EN1171
- Flange standard: EN1092-2
- Temperature range: 0 ~ 80 °C

Dimensions		Pressure rating	Size (mm)				
DN	inch	PN	ΦD	L	H1	H	Φd
50	2	16	165	150	256	338.5	203
		25					
65	2.5	16	185	170	256	348.5	203
		25					
80	3	16	200	180	273.5	373.5	203
		25					
100	4	16	220	190	323.5	433.5	300
		25	235			441	
125	5	16	250	200	376	501	300
		25	270			511	
150	6	16	285	210	423.5	566	300
		25	300			573.5	
200	8	16	340	230	530.5	700.5	406
		25	360			710.5	
250	10	16	400	250	645	845	406
		25	425			857.5	
300	12	16	455	270	725.5	953	406
		25	485			953	

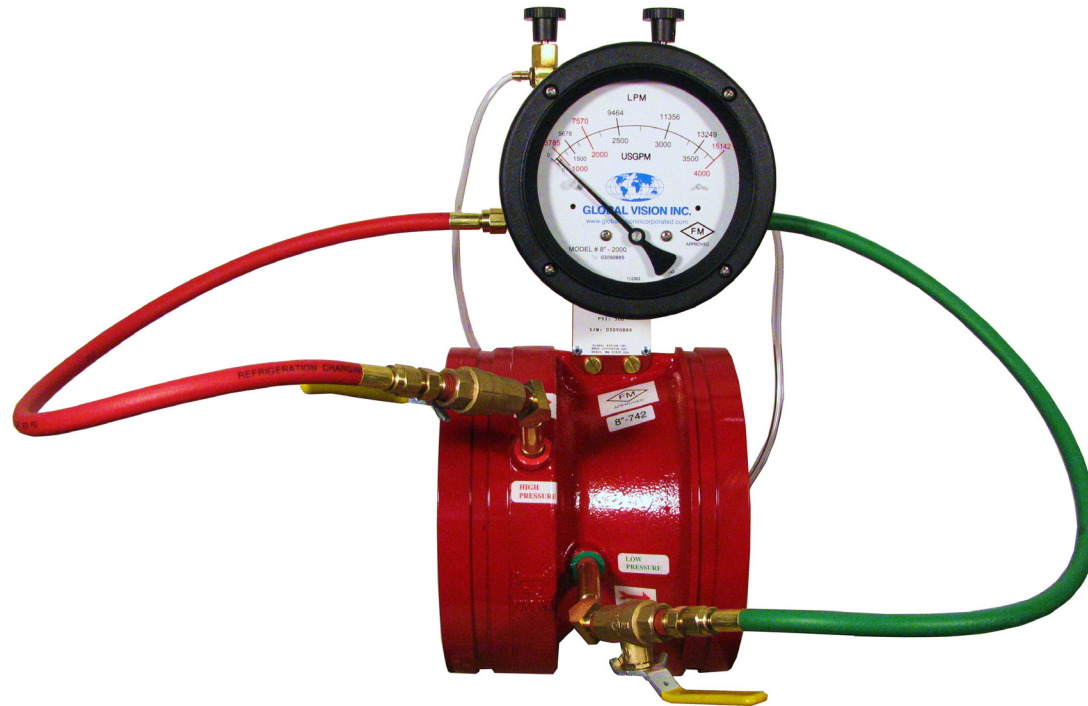


GVI

GLOBAL VISION INC

GVI FIRE PUMP TEST METERS

THE INDUSTRY LEADER IN FLOW METERS



GLOBAL VISION INC 8730 JEFFERSON HIGHWAY
OSSEO, MINNESOTA 55369 USA TEL: 763-391-0990 FAX: 763-391-7668
WWW.GVIFLOW.COM INFO@GVIFLOW.COM



WHY CHOOSE GVI?



- FM / UL LISTED FLOW METERS
- PATENTED ACCURACY OF 0.5%
- FRICTION LOSS LESS THAN 1 PSI
- 612 WARRANTY: 6YR GAUGE / 12YR VENTURI
- DIGITAL GAUGE WITH 4-20MA TRANSMITTER
- SUBSTANTIAL INVENTORY FOR QUICK SHIPPING

WITH CONTINUOUS IMPROVEMENTS AND CUTTING EDGE TECHNOLOGY, GVI HAS JUST RELEASED OUR MOST ACCURATE FLOW METER EVER.

THIS IS WHY GVI IS THE MOST REQUESTED AND MOST SPECIFIED FIRE METER IN THE WORLD.

THE GVI GAUGE IS USED IN MORE FIRE TRAINING CENTERS WORLDWIDE THAN ALL FLOW TEST METER COMPETITORS COMBINED.

AFTER USING GVI FLOW METER FOR THE FIRST TIME, YOU WILL UNDERSTAND WHY WE ARE NUMBER 1.



ACCUFLOW VENTURI
PATENT
#D764,630S



CUSTOMIZED AND SPECIAL PROJECTS



GVI HOLDS FM AND UL LISTING



DIGITAL 4-20MA TRANSMITTER



THE FIRE PUMP FLOW TEST HAS NEVER BEEN EASIER. THE NEW FM/UL APPROVED DIGITAL GAUGE FROM GVI ALLOWS USERS TO EASILY READ THE DISPLAY.

- ◆ TEMP: 32F - 140F
- ◆ POWER: 12-32 VDC
- ◆ 2 WIRE 4-20 MA
- ◆ BACKLIT DISPLAY
- ◆ REVERSE POLARITY PROTECTION
- ◆ OPTIONAL BATTERY PACK AVAILABLE

TRANSMITTING TO A COMPATIBLE CONTROLLER WILL ALSO ALLOW OPTIONS FOR CAPTURING FLOW CURVE AND ADDITIONAL DATA.



DISPLAY UNITS IN: GPM / LPM / M3H

CALIBRATION:

THE UNIT IS FACTORY CALIBRATED AND HAS NO USER ADJUSTMENTS FOR CALIBRATION. THE ASSEMBLY USES A MICROPROCESSOR TO PERFORM AN 11 POINT SQUARE ROOT LINEARIZATION ALGORITHM.

PRODUCTION / TESTING

GVI PRODUCES ALL VENTURIS FROM DOMESTIC U.S. THICK WALL TUBE. THIS HEAVY DUTY STEEL ENSURES OUR VENTURI WILL BE AROUND YEARS LONGER THAN THE THINNER, LIGHTWEIGHT ALTERNATIVES, SO MUCH LONGER, GVI HAS A 12 YEAR WARRANTY ON THIS VENTURI.



WITH 2
NEW CNC
LATHES AT
GVI, LEAD
TIMES ARE
IN HOURS,
NOT WEEKS



TOLERANCES
OF .005
MEAN A
PERFECT
VENTURI
EVERY TIME



AFTER MACHINING,
EVERY VENTURI IS
INSPECTED WITH
ISO CERTIFIED
TEST EQUIPMENT



NIST METER TESTING

EVERY GVI GAUGE UNDERGOES ADVANCED COMPUTERIZED TESTING BEFORE LEAVING THE FACTORY. THESE TESTS INCLUDE LEAKAGE, CALIBRATION, AND ACCURACY TESTS.

UPON SUCCESSFUL TESTING THE GAUGE IS ISSUED AN NIST CERTIFICATE TIED SPECIFICALLY TO THE SERIAL NUMBER ON THE DIAL FACE.

THE CERTIFICATION IS STORED IN A PLASTIC POUCH CONNECTED TO THE GAUGE. THIS POUCH ALSO ILLUSTRATES THE TYPICAL INSTALLATION AND THE OPERATING INSTRUCTIONS FOR THE FIRE PUMP FLOW TEST.

CONTACT YOUR GVI REP FOR DUPLICATE SHEETS OR UPDATED CERTIFICATIONS.



Line Size	Pump GPM	Meter Range Min. & Max. GPM	Meter Range Min. & Max. LPM	Model Number Grooved	Model Number Butt Weld	Model Number 150# Flanged	Model Number <Threaded> *300# FLANGED*
**1-1/4"	25	12.5 – 50	47 - 189	* 1.25"-25-G	1.25"-25-B	* 1.25"-25-F	1.25"-25-T
**2"	50	25 – 100	95 - 379	* 2"-50-G	2"-50-B	* 2"-50-F	2"-50-T
**2-1/2"	100	50 - 200	189 - 757	2.5"-100-G	2.5"-100-B	2.5"-100-F	*2.5"-100-F3
3"	150	75 – 300	284 - 1136	3"-150-G	3"-150-B	3"-150-F	*3"-150-F3
3"	200	100 – 400	379 - 1514	3"-200-G	3"-200-B	3"-200-F	*3"-150-F3
4"	250	125 – 500	473 - 1893	4"-250-G	4"-250-B	4"-250-F	*4"-250-F3
4"	300	150 – 600	568 - 2271	4"-300-G	4"-300-B	4"-300-F	*4"-300-F3
4"	400	200 – 800	757 - 3028	4"-400-G	4"-400-B	4"-400-F	*4"-400-F3
4"	450	225 – 900	852 - 3407	4"-450-G	4"-450-B	4"-450-F	*4"-450-F3
5"	450	225 – 900	852 - 3407	5"-450-G	5"-450-B	5"-450-F	*5"-450-F3
5"	500	250 – 1000	946 - 3785	5"-500-G	5"-500-B	5"-500-F	*5"-500-F3
5"	750	375 – 1500	1420 - 5678	5"-750-G	5"-750-B	5"-750-F	*5"-750-F3
6"	500	250 – 1000	946 - 3785	6"-500-G	6"-500-B	6"-500-F	6"-500-F3
6"	750	375 – 1500	1420 - 5678	6"-750-G	6"-750-B	6"-750-F	6"-750-F3
6"	1000	500 – 2000	1893 - 7570	6"-1000-G	6"-1000-B	6"-1000-F	6"-1000-F3
6"	1250	625 - 2500	2366 - 9462	6"-1250-G	6"-1250-B	6"-1250-F	6"-1250-F3

STANDARD MODEL LIST



Line Size	Pump GPM	Meter Range Min. & Max. GPM	Meter Range Min. & Max. LPM	Model Number Grooved	Model Number Butt Weld	Model Number Flanged
8"	1000	500 – 2000	1893 - 7570	8"-1000-G	8"-1000-B	8"-1000-F
8"	1250	625 – 2500	2366 - 9462	8"-1250-G	8"-1250-B	8"-1250-F
8"	1500	750 – 3000	2839 - 11356	8"-1500-G	8"-1500-B	8"-1500-F
8"	2000	1000 – 4000	3785 - 15142	8"-2000-G	8"-2000-B	8"-2000-F
8"	2500	1250 – 5000	4731 - 18925	8"-2500-G	8"-2500-B	8"-2500-F
8"	3000	1500 – 6000	5678 - 22712	8"-3000-G	8"-3000-B	8"-3000-F
10"	1500	750 – 3000	2839 - 11356	10"-1500-G	10"-1500-B	10"-1500-F
10"	2000	1000 – 4000	3785 - 15142	10"-2000-G	10"-2000-B	10"-2000-F
10"	2500	1250 – 5000	4731 - 18925	10"-2500-G	10"-2500-B	10"-2500-F
10"	3000	1500 – 6000	5678 - 22712	10"-3000-G	10"-3000-B	10"-3000-F
10"	3500	1750 – 7000	6624 - 26495	10"-3500-G	10"-3500-B	10"-3500-F
10"	4000	2000 – 8000	7570 - 30283	10"-4000-G	10"-4000-B	10"-4000-F
10"	4500	2250 – 9000	8518 - 34068	10"-4500-G	10"-4500-B	10"-4500-F
12"	2500	1250 – 5000	4731 - 18925	* 12"-2500-G	* 12"-2500-B	* 12"-2500-F
12"	3000	1500 – 6000	5678 - 22712	* 12"-3000-G	* 12"-3000-B	* 12"-3000-F
12"	3500	1750 – 7000	6624 - 26495	* 12"-3500-G	* 12"-3500-B	* 12"-3500-F
12"	4000	2000 – 8000	7570 - 30283	* 12"-4000-G	* 12"-4000-B	* 12"-4000-F
12"	4500	2250 – 9000	8518 - 34068	* 12"-4500-G	* 12"-4500-B	* 12"-4500-F
12"	5000	2500 - 10000	9464 - 37854	12"-5000-G	12"-5000-B	12"-5000-F



**1026
Carbon Steel**

316 stainless

Duplex S.S.

Super Duplex

Monel

NiAlumBrz

90-10 Cu-Ni

Aluminum

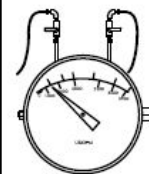
Brass

Epoxy

CONSTRUCTION & SPECIFICATIONS

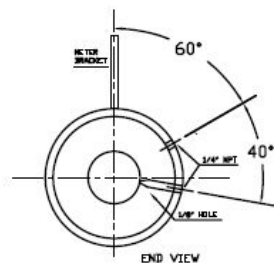
MATERIALS: 1026 MACHINED CARBON STEEL
VALVES: 1/4" BRASS BALL
FLANGE OPT: 150# ANSI B 16.5
300# ANSI B 16.5
ID TAG: ALUMINUM

METER DATA

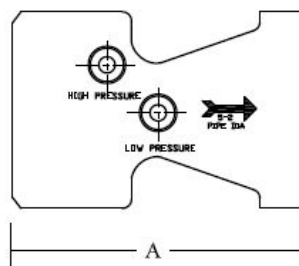


CONSTRUCTION: ALUMINUM BODY
OPERATION: DIAPHRAGM (BUNA CONVOLUTED)
TEMPERATURE: 180°F / 80°C
PRESSURE: 500 PSIG - 3450 KPA
APPROX. WEIGHT: 3.75 LBS (1.7 KGS)

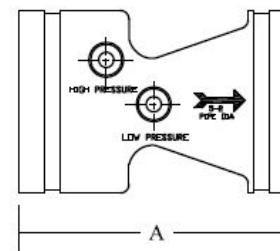
4" DIAL STANDARD - 6" DIAL AVAILABLE



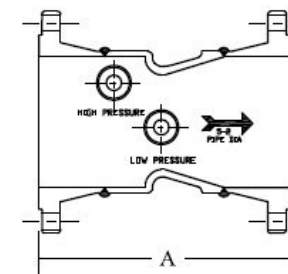
BUTT WELD / GROOVED



BUTT WELD



GROOVED



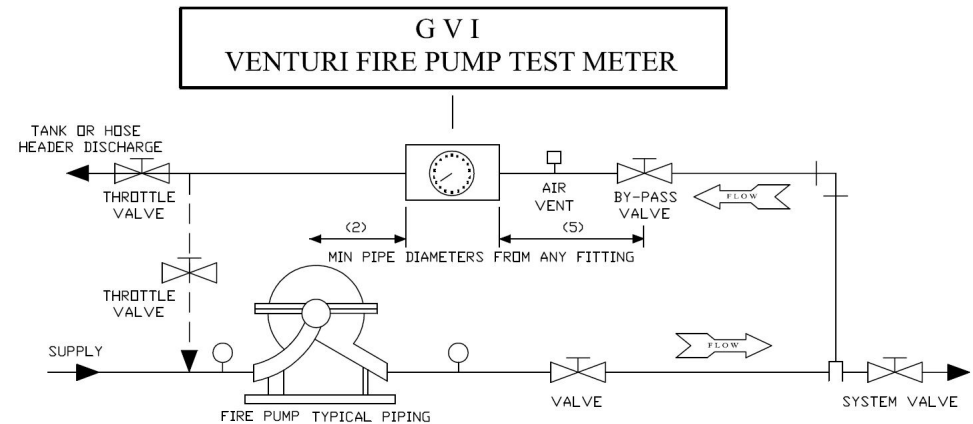
FLANGED

SIZE	BUTT WELD & GROOVED		150 # FLANGED		300# FLANGED	
	"A" DIM	APPROX. WEIGHT	"A" DIM	APPROX. WEIGHT	"A" DIM	APPROX. WEIGHT
1-1/4"	4.875	10 lbs.	—	—	—	—
2"	5.000	11 lbs.	—	25 lbs.	—	—
2-1/2"	4.000	11 lbs.	9.250	25 lbs.	10.000	35 lbs.
3"	4.000	13 lbs.	9.375	40 lbs.	10.250	50 lbs.
4"	5.375	15 lbs.	11.375	50 lbs.	12.125	65 lbs.
5"	6.000	20 lbs.	13.000	60 lbs.	13.750	80 lbs.
6"	7.000	25 lbs.	14.000	75 lbs.	14.750	115 lbs.
8"	7.250	35 lbs.	15.250	120 lbs.	16.000	175 lbs.
10"	8.000	60 lbs.	16.000	170 lbs.	17.250	225 lbs.
12"	10.000	65 lbs.	19.000	225 lbs.	20.250	345 lbs.

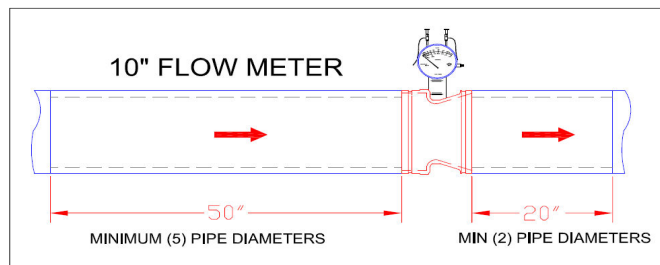
INSTALLATION / OPERATION



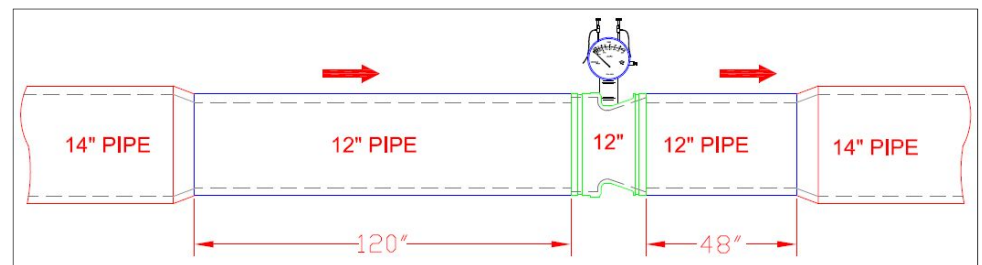
1. CLOSE THE SYSTEM VALVE
2. OPEN THE BY-PASS VALVE AND THROTTLE VALVE
3. PURGE AIR FROM METER AS FOLLOWS:
 - OPEN BALL VALVE ON RED AND GREEN HOSES
 - OPEN THE HIGH SIDE BLEEDER KNOB UNTIL A STEADY STREAM OF WATER IS COMING FROM THE CLEAR HOSE
 - CLOSE HIGH PRESSURE BLEEDER KNOB
 - REPEAT FOR LOW PRESSURE SIDE
 - ENSURE BOTH BLEEDER KNOBS ARE CLOSED
4. START FIRE PUMP, READ METER IN GPM/LPM/M3H
5. REFER TO PUMP REQUIREMENTS AND ADJUST THROTTLE VALVE FOR THIS REQUIREMENT
6. AFTER TEST, OPEN SYSTEM VALVE AND CLOSE THE BY-PASS AND THROTTLE VALVES



TYPICAL FLOW LOOP CONFIGURATION
CAN BE INSTALLED VERTICALLY OR HORIZONTALLY



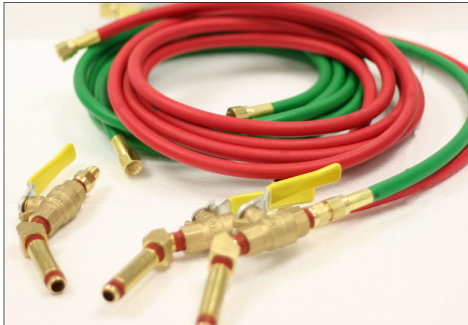
TYP. INSTALL WITH 5/2 PIPE DIAMETERS.
CLEAR OF OBSTRUCTIONS, VALVES, OR BENDS.



IF REDUCERS ARE USED, LENGTHEN THE MIN PIPE DIA.
TO 10/4 TO ENSURE ACCURACY IS MAINTAINED.



REPLACEMENT HOSES



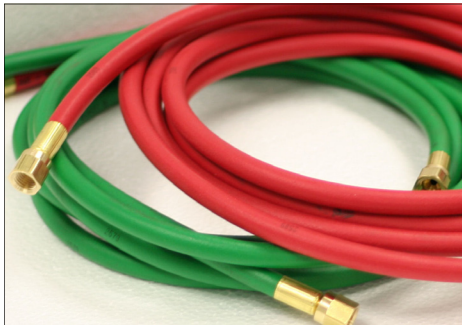
Replacement Hoses with Full Brass Fittings

GREEN HOSE

24-FPH-F-GRN	2' Green fire pump hose with brass fittings
36-FPH-F-GRN	3' Green fire pump hose with brass fittings
60-FPH-F-GRN	5' Green fire pump hose with brass fittings
120-FPH-F-GRN	10' Green fire pump hose with brass fittings
180-FPH-F-GRN	15' Green fire pump hose with brass fittings

RED HOSE

24-FPH-F-RED	2' Red fire pump hose with brass fittings
36-FPH-F-RED	3' Red fire pump hose with brass fittings
60-FPH-F-RED	5' Red fire pump hose with brass fittings
120-FPH-F-RED	10' Red fire pump hose with brass fittings
180-FPH-F-RED	15' Red fire pump hose with brass fittings



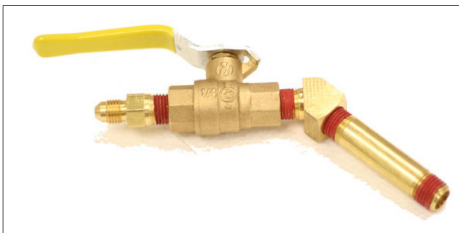
Replacement Hoses WITHOUT Brass Fittings

GREEN HOSE

24-FPH-GRN	2' Green fire pump hose only
36-FPH-GRN	3' Green fire pump hose only
60-FPH-GRN	5' Green fire pump hose only
120-FPH-GRN	10' Green fire pump hose only
180-FPH-GRN	15' Green fire pump hose only

RED HOSE

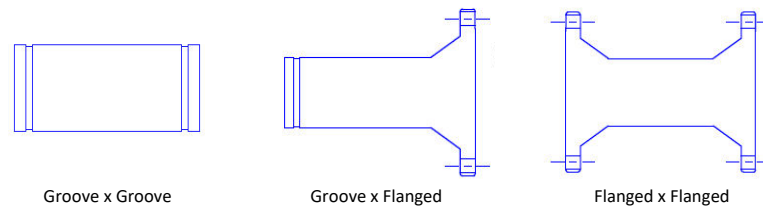
24-FPH-RED	2' Red fire pump hose only
36-FPH-RED	3' Red fire pump hose only
60-FPH-RED	5' Red fire pump hose only
120-FPH-RED	10' Red fire pump hose only
180-FPH-RED	15' Red fire pump hose only



Brass Fitting Assembly

BFA1 - Single assembly
BFA2 - Brass fitting pair

GVI Spooling Pieces



FLOW METER ACCESSORIES



CONTACT YOUR GVI REPRESENTATIVE FOR ADDITIONAL
INFORMATION ON REPLACEMENT PARTS AND ACCESSORIES.



ANALOG GAUGE



DIGITAL GAUGE



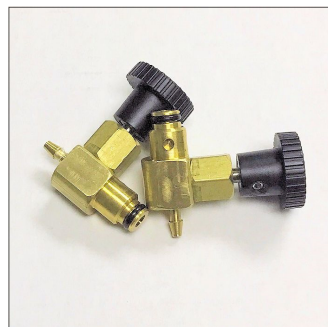
DIGITAL BATTERY PACK



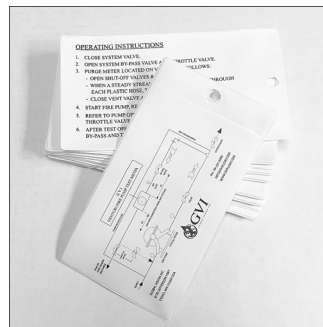
HARD STORAGE CASE



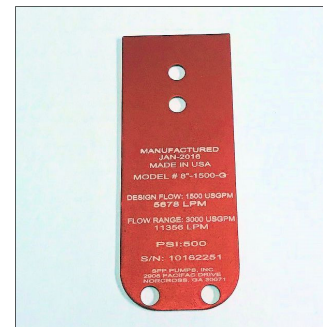
CLEAR BLEED HOSE



BLEEDER VALVES



INSTRUCTION POUCH



ENGRAVED I.D. TAG



AUTO AIR VENT



FACEBOOK



GVIFLOW.COM



GLOBAL VISION INC 8730 JEFFERSON HIGHWAY
OSSEO, MINNESOTA 55369 USA
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Reliable®

Miscellaneous System Components

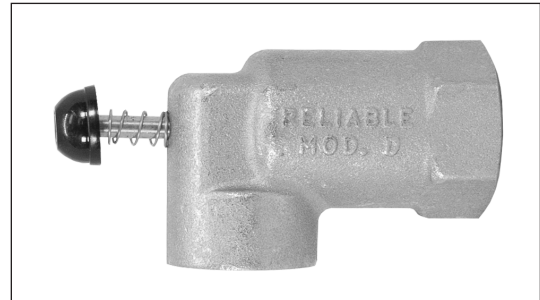
Model C — Automatic Ball Drip

An automatic drain valve horizontally installed at the low point in the fire department connection piping of automatic sprinkler systems. Water pressure from a fire department pumper automatically closes this valve. It automatically re-opens when pressure ceases, permitting this piping to drain and thereby preventing freezing. Made of bronze and available with $\frac{3}{4}$ " (R $\frac{3}{4}$) or $\frac{1}{2}$ " NPT (R $\frac{1}{2}$) female inlet connection. FM approved. Length: 2 $\frac{9}{16}$ ". Maximum working pressure: 175psi (12bar).



Model D - Mechanical Ball Check Valve

The Model D Mechanical Ball Drip Valve is a trim component used in the alarm line of various Reliable valves. The mechanical ball drip valve is designed to move to the closed position upon activation of the dry or deluge valve. After valve activation, push in the plunger of the mechanical ball drip valve to manually release the water pressure and to drain the alarm line. Made of bronze and available with $\frac{1}{2}$ " NPT (R $\frac{1}{2}$) female inlet connection. Length: 3 $\frac{1}{2}$ ". Listed/ Approved as a trim component. See individual trim bulletins for details.



Model D - Semi-Automatic Ball Check Valve

The Model D Semi-automatic Ball Check Valve is a listed trim component used in the alarm line of various Reliable dry pipe and deluge valves. In the normal (open) position the semi-automatic ball drip allows for relieving small amounts of water in the intermediate alarm chamber of the valve. The semi-automatic ball check valve is designed to close upon activation of the dry or deluge valve when sufficient flow is present in the alarm line. When resetting the valve, push in the plunger of the semi-automatic ball check valve to manually release the water pressure and to drain the alarm line. Made of bronze and available with $\frac{1}{2}$ " NPT (R $\frac{1}{2}$) female inlet and outlet connections. Approximate length: 3 $\frac{1}{2}$ ". Listed/ Approved as a trim component. See individual trim bulletins for details.



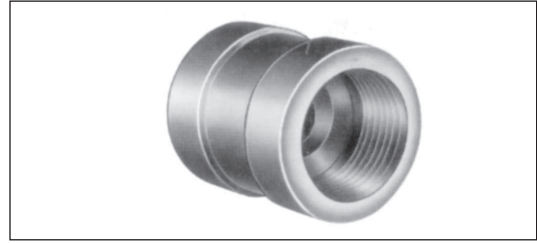
Model A — Fill Cup

Made of cast iron. Available with $\frac{1}{2}$ " or $\frac{3}{4}$ " NPT (R $\frac{1}{2}$ or R $\frac{3}{4}$) female pipe connection. Cup Diameter: 3 $\frac{3}{4}$ " (95mm). Length: 2 $\frac{1}{4}$ " (57mm).



Model A — Blind Test Connection

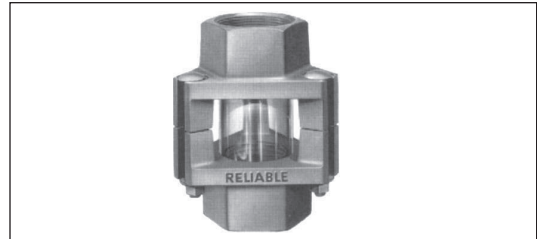
Designed for installation in test lines of sprinkler systems that connect to open drains. Made of bronze with 1" NPT female pipe connections. Orifice is sized to simulate the flow rate equivalent to one nominal 1/2" (15mm) orifice sprinkler.
Length: 17/8" (48mm).



Model B — Sight Test Connection

Designed for installation on the drain side of the test valve in a test line that connects to a closed drain.

Made of cast iron with clear tube. Smooth bore non-corrosive orifice gives flow equivalent to one nominal 1/2" (15mm) orifice sprinkler. Has 1" NPT pipe connections.
Length: 51/16" (129mm).



Model UA Water Pressure Gauge

Range 0 to 300psi in 5psi increments, and 0 to 2000 kPa in 50kPa increments. 1/4" NPT (R1/4) male pipe connection. Case: 33/4" diameter (95mm). Height: 43/4" (121mm). Accuracy: ANSI B40.1 Grade B (3–2–3%)
Underwriters Laboratories Listed, UL file EX26795
Factory Mutual Approved Std. Class No. 2311



Model UA Air Pressure Gauge

Range 0 to 80psi in 1psi increments, and 0 to 550kPa in 10kPa increments. Retard to 250psi and 1750kPa. 1/4" NPT (R1/4) male pipe connection. Case: 33/4" diameter (95mm). Height: 43/4" (121mm). Accuracy: ANSI B40.1 Grade B (3–2–3%)
Underwriters Laboratories Listed, UL file EX26795
Factory Mutual Approved Std. Class No. 2311



Model UA Air-Water Pressure Gauge

Range 0 to 300psi in 5psi increments, and 0 to 2000 kPa in 50kPa increments. 1/4" NPT (R1/4) male pipe connection. Case: 33/4" diameter (95mm). Height: 43/4" (121mm). Accuracy: ANSI B40.1 Grade B (3–2–3%)
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